Robust ECG Biometrics by Fusing Temporal and Cepstral Information

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Introduction

The ECG signal is an emerging novel behavior biometric for human identification.

- Individual differences in the heart structure manifest unique characteristics in their ECG signals which can be used as a biometric trait.
- The ECG signal can provide real-time liveness feedback.
- The usage of vital signs is a potentially viable approach in a variety of application scenarios such as security and personalized health care.

One disadvantage is that the ECG signal inherently varies at different heartbeats of the same subject due to

- variations in fitness, physical and emotional states
- variabilities caused by sensor position changes
- long term baseline shifts.

Compared to systems using ECG signals from multiple leads [1], promising results based on a single lead ECG signal have also been proposed in [2, 3, 4, 5, 6].

Therefore, in this paper, our focus is on ECG biometrics for healthy people from a single lead sequential signal under rest condition.

Conventional temporal domain approaches:

1. the reduced feature sets derived from ECG characteristics points [2, 3, 4, 5] plus supervised classification
   the fiducial points extraction is not robust for all possible types of ECG traces and only a certain percentage of the population can be successfully enrolled [6].
2. the principal component analysis (PCA) approach at the appearance level [5, 6]
   The PCA approach is a kind of generative modeling and requires model scoring on every normalized heart beat in a long sequence of heartbeats.
3. long term (5-seconds window) autocorrelation in conjunction with discrete cosine transform as feature extraction plus nearest neighbor matching [5]
   The ECG signal within the long processing window is non-stationary, it may negatively influence the system performance.

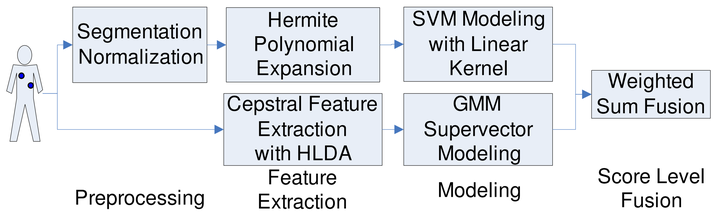
The first two approaches require ECG signal pre-processing (peak detection, heartbeat segmentation and normalization) which can be computationally expensive.

The proposed hybrid approach by fusing temporal and cepstral information:

1. In temporal domain, after ECG signal pre-processing, Hermite polynomial expansion (HPE) and SVM are used to capture the ECG shape characteristics and perform supervised classification, respectively.
   - Hermite polynomials are a classical orthogonal polynomial sequences and have been successfully used to describe the ECG signal [7].
   - SVM is a discriminative modeling for classification.
   - The HPE projection matrix and linear kernel SVM scoring functions are combined together into a model vector, which makes the HPE feature extraction and sequential scoring just one simple inner product for computational efficiency.
2. In cepstral domain, Gaussian Mixture Models (GMM) is employed to model the cepstral features.
   - No need for ECG signal pre-processing.
   - Cepstral features are especially conducive for mitigating convolutional effects.
   - ECG Cepstral features have been successfully adopted in our previous work for physical activity recognition [8] .
   - Dimension reduction by Heteroscedastic Linear Discriminant Analysis (HLDA) [9].
   - Discriminative modeling via GMM Supervector (GSV) kernel [10].
3. Fusion of both temporal and cepstral system outcomes at the score level is used to improve the overall performance.
   - Temporal and cepstral information are complementary.
   - Weighted sum fusion strategy is adopted.

Methods

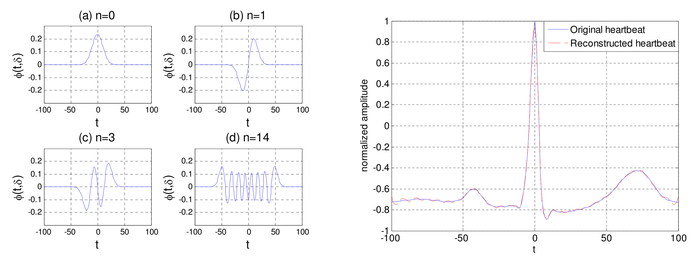
System overview



ECG signal pre-processing

Each heartbeat waveform is normalized to the same time scale and amplitude scale by using standard methods in the ECG toolbox [6, 11].

Hermite polynomial expansion



Let us denote the ECG curve vector and polynomial order by $a(t)$ and L respectively. The Hermite polynomial expansion can be written as follows [7]:

$$a(t) = \sum_{n=0}^{L-1} c_n \phi_n(t, \delta) \qquad t \in [-M, M] \tag{1}$$

where c_n $(n = 0, \cdots, L-1)$ are the HPE coefficients and $\phi_n(t, \delta)$ is the Hermite basis function defined as

$$\phi_n(t, \delta) = \frac{1}{\sqrt{\delta 2^n n! \sqrt{\pi}}} e^{-t^2/2\delta^2} H_n(t/\delta) \tag{2}$$

The functions $H_n(t/\delta)$ are the physicists' Hermite polynomials which are defined recursively by:

$$H_0(t) = 1, H_1(t) = 2t \tag{3}$$

$$H_n(t) = 2tH_{n-1}(t) - 2(n-1)H_{n-2}(t) \tag{4}$$

Cepstral feature extraction

- Bandpass pre-filtering (0.05-40 HZ FIR) to reduce both high frequency noises and low frequency baseline fluctuations
- 32 cepstral coefficients from 32 frequency band filter on 500 msec segments with 50% overlap, first order delta features included
- HLDA [9] used for dimension reduction

GMM and GMM Supervector (GSV) modeling

Universal Background Model (UBM) using a Maximum A Posteriori (MAP) adaptation approach [12] is used.

The GSV approach consists of support vector machines with GMM Supervectors, which is a concatenation of the GMM mean vectors, as input features for the SVM. A linear kernel based upon an approximation to KL divergence between two GMM models was derived in [10] as follows:

$$K(\mu, \hat{\mu}) = \sum_{i=1}^{M} (\sqrt{p_i}\Sigma_i^{-\frac{1}{2}}\mu_i)(\sqrt{p_i}\Sigma_i^{-\frac{1}{2}}\hat{\mu}_i)^t \tag{5}$$

where p_i and $\sum_i$ are the ith UBM mixture weights and diagonal covariance matrix; μ_i corresponds to the mean of the ith Gaussian component in this GMM.

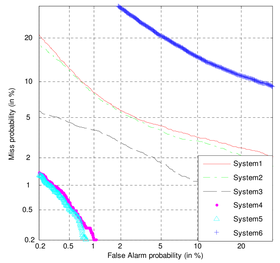
Experimental results

MIT-BIH Normal Sinus Rhythm Database [13], 18 healthy subjects. For each subject, data from hour 1-2 and hour 2-3 are used as training data and testing data, respectively. The experimental results in this paper are based on a 10 seconds testing duration.

Performance in close-set identification accuracy P_c and Equal Error Rate (EER):

System ID	Pre-Process	Methods	P_c (%)	EER (%)
1	no	Cepstral: GMM	93.96	4.30
2	no	Cepstral: GMM+HLDA	94.78	4.05
3	no	Cepstral: GSV+HLDA	95.90	2.5
4	yes	Temporal: HPE+SVM	98.11	0.55
5	yes	Fusion of system (3+4)	98.26	0.50
6	yes	EigenPulse: PCA [6]	89.46	13.2

Performance in Detection Error Tradeoff (DET) curves:



Conclusion and future work

- In the time domain, HPE and SVM are used to efficiently model the intra-heartbeat patterns of different individuals with linear kernel.
- In the frequency domain, without the ECG pre-processing, cepstral features combined with GMMs directly model the short time characteristics of the ECG signal.
- By fusing both temporal and cepstral information together, the overall biometric system performance is improved.
- Future work would include validating the performance over a much larger population of test subjects, comparing/combining with other biometrics modalities, and investigating the robustness against a variety of physical and emotional state variabilities. Ongoing data collection: http://knowme.usc.edu/experiments.html

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